What is JSX?

JSX stands for: JavaScript XML

It allows us to write **HTML-like code inside JavaScript**.

Simple Definition

JSX is a syntax used in React to create UI easily.

Instead of writing complex JavaScript, we can write code that looks like HTML.

Without JSX

```
const element = React.createElement(
  "h1",
  null,
  "Hello React"
);
```

This is difficult to read.

With JSX

```
const element = <h1>Hello React</h1>;
```

Why JSX is Used?

JSX helps developers to:

- Write cleaner code
- Create UI easily
- Combine HTML + JavaScript together
- Improve readability

JSX Looks Like HTML

```
function App() {
  return (
    <div>
      <h1>Welcome</h1>
      <p>Learning JSX</p>
    </div>
  );
}
```

Important Point

JSX is NOT actual HTML.

Browser cannot understand JSX directly.

React converts JSX into JavaScript using tools like Babel.

```
JSX Code
   ↓
Babel Converts
   ↓
JavaScript
   ↓
Browser Displays UI
```

Rule 1: Return One Parent Element

Wrong

```
return (
  <h1>Hello</h1>
  <p>World</p>
);
```

Correct

```
return (
  <div>
    <h1>Hello</h1>
    <p>World</p>
  </div>
);
```

Attributes and Comments in React

In React, attributes and comments are used almost like HTML, but with some React rules.

1. Attributes in React

What are Attributes?

Attributes provide extra information to HTML elements.

Example:

- `src`
- `href`
- `alt`
- `type`
- `value`

In React, attributes are written inside JSX.

```
function App() {
  return (
    <img
      src="image.png"
      alt="Profile"
    />
  );
}
```

Rule 1: Use `className` Instead of `class`

Because `class` is reserved keyword in JavaScript.

Wrong

```
<div class="box"></div>
```

Correct

```
<div className="box"></div>
```

Common React Attributes

Attribute	Use
`className`	CSS class
`src`	Image source
`href`	Link
`alt`	Alternative text
`type`	Input type
`value`	Input value
`placeholder`	Input hint
`onClick`	Click event

Example Using Multiple Attributes

```
function App() {
  return (
    <a
      href="https://google.com"
      target="_blank"
    >
      Google
    </a>
  );
}
```

```
function App() {
  const image = "photo.png";

  return <img src={image} alt="Profile" />;
}
```